Work Breakdown Structure (WBS)

Project: ResiTrack — Apartment / Residencies Organizer
Team: H14
Course: Software Engineering

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1. Purpose

Breaks the ResiTrack mini-project into work packages, each with a deliverable, an owner and a timeline. Implementation packages map to the GitHub backlog issues (issue numbers in brackets) and to the two sprints planned on the project board.

2. Team

Name	SRN
Shreesh Athreya	PES1UG24CS441
Shree Krishna	PES1UG24CS440
Swaroop Venkateshwar	PES1UG24CS923
Sushanth S Rao	PES1UG24CS485

3. Work Breakdown

ID	Work Package	Deliverable / Output	Owner	Timeline
1	**Project Management**			
1.1	Team formation and repository setup	GitHub repo with submission folder structure	Swaroop Venkateshwar	13 Oct
1.2	GitHub Project and product backlog	29 backlog items from SRS FRs/NFRs, with story points and owners	All	12–13 Oct
1.3	Sprint planning	Sprint 1 (54 pts) and Sprint 2 (46 pts) iterations on the project board	All	13 Oct
2	**Requirements Engineering**			
2.1	Requirements elicitation and specification	SRS (IEEE 830): 15 FRs, 10 NFRs, 6 security requirements	All	by 28 Sept
2.2	Use case modelling	UML use case diagram, 15 use cases, 5 actors	All	by 28 Sept
2.3	Traceability	Requirements Traceability Matrix (SRS §6)	All	by 28 Sept
2.4	Requirements process and change management	RE process document (3-RE)	Shreesh Athreya	13 Oct

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3	**Architecture and Design**			
3.1	Architecture pattern and components	3-tier client–server with MVC server; component diagram	All	by 28 Sept
3.2	Detailed design	UML sequence diagrams, REST API design, error handling, data design	All	by 28 Sept
3.3	Security architecture	JWT auth, RBAC middleware, bcrypt, TLS, audit logging design	All	by 28 Sept
4	**Test Planning**			
4.1	Test plan (IEEE 829)	Scope, approach, security validation, plan elements	All	by 28 Sept
4.2	Test case design	25 test cases traced to every FR and NFR (rev. 1.1)	All	13 Oct
5	**Implementation – Sprint 1**			19–23 Oct
5.1	Project scaffold (MERN, MVC) and GitHub Actions CI	Issues #1, #2	Sushanth S Rao, Swaroop Venkateshwar	Sprint 1
5.2	Authentication and security	Login/JWT, RBAC, bcrypt, session timeout (#3–#6)	Swaroop Venkateshwar	Sprint 1
5.3	Tenant, flat and rent management	Flat inventory, tenant registration, rent payments (#7–#9)	Sushanth S Rao	Sprint 1
5.4	Notices and maintenance requests	Notices, raise request, update status (#10–#12)	Shree Krishna	Sprint 1
5.5	Visitors, complaints and audit log	Visitor log, complaints, admin audit log (#13–#15)	Shreesh Athreya	Sprint 1
6	**Implementation – Sprint 2**			26–30 Oct
6.1	Rent reminders and PDF receipts	node-cron reminders, receipt generation (#17, #18)	Sushanth S Rao	Sprint 2
6.2	Document upload	PDF/JPEG ≤5 MB upload with validation (#21)	Swaroop Venkateshwar	Sprint 2
6.3	Kanban maintenance dashboard	Drag-and-drop status board (#24)	Shree Krishna	Sprint 2
6.4	Owner financial summary	Rent collected, dues, occupancy report (#28)	Shreesh Athreya	Sprint 2
7	**Testing and Quality**			
7.1	Unit and integration tests	≥60% coverage, MVC review (#29)	Shreesh Athreya	Sprint 2
7.2	Performance and scalability tests	Load test at 200 users; 500-record test (#22, #23)	Swaroop Venkateshwar	Sprint 2
7.3	Usability and responsive UI tests	375px layout checks; timed admin onboarding (#25, #26)	Shree Krishna	Sprint 2

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7.4	Security validation pass	Test Plan §5.1 checks run before final submission	All	Sprint 2
8	**Deployment**			
8.1	Hosting, TLS and uptime monitoring	Deployed client/server, HTTPS, uptime monitor (#19, #20)	Sushanth S Rao	Sprint 2
9	**Documentation and Demo**			
9.1	Sprint 1 demo video	1-minute video of working product in 9-Demo (#16)	Swaroop Venkateshwar	23 Oct
9.2	Sprint 2 demo video and code freeze	2-minute video; final documentation; development frozen (#27)	Shree Krishna	30 Oct
9.3	Deviation notes	Changes from submitted SRS/SAD/Test Plan documented for the final demo	All	30 Oct

4. Milestones

Milestone	Date	Output
Part-1 documents	28 Sept 2026	SRS, Test Plan, Architecture & Design Specification
Backlog creation	12 Oct 2026	GitHub Project with backlog, story points, owners
Sprint 1	19–23 Oct 2026	Working core product; 1-minute demo video
Sprint 2	26–30 Oct 2026	Complete product; 2-minute demo video; documentation; code freeze